

Vaishnavi Mocherla

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EDUCATION

San Jose State University <i>M.S. in Applied Data Science</i>	San Jose, CA <i>Aug 2023 – May 2025</i>
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EXPERIENCE

Informatica <i>Data Engineer – AI & Cloud Systems</i> – Developed a AI-powered analytics agent using Azure OpenAI and Python, enabling NL-to-SQL query translation and reducing analyst workload by 60%; improved SQL accuracy by 20% via prompt tuning and schema mapping. – Built scalable ETL pipelines with Informatica IICS to ingest structured/unstructured data, and applied IDQ validation to reach 99.9% data accuracy. – Engineered Databricks pipelines with PySpark + SQL to optimize real-time and batch processing by 35%; containerized workflows with CI/CD for production reliability. – Configured and managed EMR Clusters to run Spark jobs on AWS, optimizing big data workloads with 30% faster job completion and reduced compute costs via spot instance tuning. – Enforced row-level security and data masking via Informatica CDAM during 1TB+ Azure Data Lake migration, ensuring full compliance with enterprise governance. – Built interactive Power BI dashboards for marketing and ops teams, reducing report turnaround time and increasing visibility into campaign performance.	California, USA <i>May 2024 – Dec 2024</i>
Kanerika Software Pvt Ltd <i>Data Engineer & ML Research</i> – Designed and delivered a BERT-based system to assess resume-JD fit and predict candidate progression up to L3, reducing manual screening by 60% across 100K+ resumes, and enabling early filtering of 35–40% of low-fit profiles. – Built a custom parser to process 10+ resume formats (PDF, DOCX, scanned text), improving extraction reliability by 70%, and integrated it with a real-time ETL pipeline using Airflow, Spark, and Python. – Developed and deployed a prediction pipeline storing results in Azure Data Lake, with outcomes visualized via Power BI dashboards tracking model confidence, fit scores, and pipeline conversion; improved screening turnaround by 30%. – Presented prototype to stakeholders, securing beta approval and roadmap inclusion for ATS integration.	Hyderabad, India <i>Aug 2022 – Jun 2023</i>

PROJECTS

Multi-Agent LLM System for Autonomous SDLC <i>Snowflake, LangChain, Semantic Kernel, Llama 3, Pinecone, Streamlit</i> – Built an autonomous, production-style LLM system with 6 collaborating agents (PM, SE, DE, QA, Deployer, Docs) simulating the full SDLC — from user requirement to GitHub deployment and documentation. – Integrated LangChain and Semantic Kernel for agent orchestration and memory routing; deployed specialized models (Llama 3, Mistral, CodeT5) fine-tuned on role-specific datasets. – Enabled real-time task execution via Streamlit UI with Pinecone-powered RAG for long-horizon query memory; incorporated Snowflake for data-driven agent responses and SQL workflows. – Achieved Less than 2s end-to-end latency, 88% QA validation accuracy, and 40% reduction in LLM hallucinations; deployed outputs with live documentation through CI-integrated GitHub workflows.	
Cloud-Native Data Engineering Platform for Responsible AI <i>Airflow, MLflow, Neo4j, Azure, LangChain</i> – Designed and deployed an end-to-end cloud-native data pipeline using Airflow, Azure Data Lake, and MLflow, enabling automated model training, versioning, and CI/CD deployment across staging and production environments. – Integrated graph-aware feature pipelines with Neo4j and built LLM-powered data validation reports using LangChain + LLaMA3, supporting explainable workflows and Responsible AI compliance at the data layer.	

TECHNICAL SKILLS

Languages: Python, SQL, R(Familiar)
Data Engineering: Spark, Kafka, Databricks, Snowflake, Hadoop, BigQuery
ETL & Data Workflow: Informatica (IICS, IDQ, CDAM), Airflow, Azure Data Factory, CI/CD
LLMs & Generative AI: Multi-Agent Systems, LangChain, Semantic Kernel, Azure OpenAI, RAG Pipelines, Pinecone, Prompt Engineering, Fine-tuning, Vector Search
Cloud & DevOps: Azure, AWS (EMR, EC2, S3, Glue, Lambda), GCP, Docker, Kubernetes, Terraform
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, HDFS
Data Processing & Analytics: Pandas, NumPy, PySpark, Scikit-learn, TensorFlow, Time Series Forecasting
BI & Visualization: Power BI, Tableau, Looker, Matplotlib, Seaborn